

Assignment 1

Total: 35

This assignment must be handwritten. Show ALL steps in ALL questions.

1. [2.5 * 8 = 20 Marks] Perform the following base conversions
 - (a) $(10110010101)_{2s} = (?)_7$
 - (b) $(00110010101)_{1s} = (?)_9$
 - (c) $(0011)_{BCD} = (?)_5$
 - (d) $(010010101011111)_{2s} = (?)_{10}$
 - (e) $(101010101000011)_{2s} = (?)_{10}$
 - (f) $(010010101011111)_{1s} = (?)_{10}$
 - (g) $(101010101000011)_{1s} = (?)_{10}$
 - (h) $(110\ 0011)_{10} = (?)_{\text{Excess-5}}$
2. [2.5 Marks] Suppose you have two 4-bit registers where your computer stores signed numbers. Now subtract 5 from -6 using your computer's registers, and figure out whether there is any overflow or not.
Note: The computer uses 2's complement representation for signed numbers.
3. [2.5 Marks] What is the largest number you can store in a 12-bit register using a 1's complement system? Your final answer should be in Octal.
4. [2.5 Marks] What is the smallest number you can store in an 8-bit system using a sign & magnitude representation? Your final answer should be in HEX.
5. [2.5 Marks] Divide 43234212_5 by 124_5 ; Find the quotient and remainder.
6. [5 Marks] You are a computer engineer and you want to buy two 8 GB DDR4 RAMs. Each RAM costs $(1C2)_{16}$ dollars. You also want to buy a graphics card RTX4070Ti, which costs $(10010110000)_2$ dollars. However, you don't have that much money with you, and you are afraid to ask your parents about it. Suddenly, one of your generous friends agreed to give you the money you need. He decided to give you $(4064)_8$ dollars. How much will you have left after buying those components? [Show the answer in decimal]